

Merewether High School
Stage 6
Software Engineering
Year 12 Trial Examination 2025
Topic Areas- All course material



Assessment Task Information

Task: 4
Task Weighting: 30% of the Assessment Schedule for the Year 12 Software Engineering Course

Section I - [MULTIPLE CHOICE – 20 Marks]

1	D
2	A
3	C
4	B
5	C
6	A
7	B
8	C
9	C
10	A

11	C
12	D
13	C
14	B
15	C
16	D
17	B
18	B
19	A
20	A

Section II [60 Marks]

Question 21 [4 marks]

4 Marks	Students construct a fully correct structure chart for the given algorithm
3 Marks	Students construct a mostly correct structure chart for the given algorithm
2 Marks	Students construct a structure chart with some correct components for the given algorithm
1 Mark	Students attempt to create a structure chart
<pre> graph TD Root[AverageListOfNumbers] GetNumbers[GetNumbers] CalculateAverage[CalculateAverage] PrintAverageToScreen[PrintAverageToScreen] Root -- arrayOfNumbers --> GetNumbers Root -- average --> CalculateAverage CalculateAverage -- average --> Root Root -- arrayOfNumbers --> PrintAverageToScreen PrintAverageToScreen -- average --> Root </pre>	

Question 22 [6 marks]

6 Marks	Students correctly explain the three types of machine learning
4-5 Marks	Students correctly explain two types of machine learning
2-3 Marks	Students correctly describe the three types of machine learning
1 Mark	Students correctly identify types of machine learning
<i>The three main types of machine learning are supervised learning, unsupervised learning, and deep learning.</i> <i>Supervised learning occurs when an algorithm is trained on labelled data, where both the input and the correct output are provided. The system learns patterns by comparing its predictions with the known answers and adjusting to improve accuracy. This approach is commonly used in applications such as spam email detection, handwriting recognition, and medical diagnosis, where large sets of labelled examples are available.</i> <i>Unsupervised learning works with unlabelled data, meaning the algorithm must identify hidden patterns or relationships without predefined answers. It is typically used for clustering, grouping, or anomaly detection. For example, businesses use unsupervised learning to segment customers into different market groups or streaming platforms use it to suggest similar movies based on viewing history. Unlike supervised learning, there is no "correct" output provided.</i> <i>Deep learning is a specialised form of machine learning that uses artificial neural networks with many layers ("deep" networks) to process data. These networks can automatically learn features from raw inputs, making them highly effective for tasks involving complex, unstructured data such as images, speech, or video. For example, deep learning powers facial recognition systems, natural language processing (such as virtual assistants), and autonomous vehicles. Deep learning requires large datasets and significant computing power but achieves high accuracy in solving complex problems.</i>	

Question 23 [3 marks]

3 Marks	Students construct a fully correct pseudocode algorithm for the FindItem routine
2 Marks	Students construct a somewhat correct pseudocode algorithm for the FindItem routine or a fully correct routine using Python rather than pseudocode
1 Mark	Students attempt to construct a pseudocode algorithm for the FindItem routine
	<pre> BEGIN FindItem(ItemID) Found = ItemExists(ItemID) IF Found THEN Location = CheckLocation(ItemID) ENDIF RETURN Location END </pre>

Question 24 [4 marks]

4 Marks	Students correctly discuss the role and importance of programmed data backup and version control when producing and implementing code, providing definitions and examples
3 Marks	Students correctly discuss the role and importance of programmed data backup and version control
2 Marks	Students correctly identify the role and importance of programmed data backup and version control
1 Mark	Students correctly identify the role and importance of programmed data backup or version control
	<i>Programmed data backup refers to the automated process of regularly saving copies of code and project files so they can be restored if data is lost or corrupted. This is important because it protects against hardware failure, accidental deletion, or cyber-attacks, ensuring that critical work is not permanently lost. For example, cloud-based backups such as Google Drive or automated server backups allow developers to recover previous versions of source code after a system crash.</i> <i>Version control is a system that tracks changes to code over time, allowing multiple developers to collaborate while maintaining a record of modifications. Its importance lies in preventing conflicts, enabling rollbacks to earlier versions if errors occur, and supporting collaborative workflows. Tools such as Git and GitHub provide branching, merging, and commit history features that make large-scale software development efficient and reliable.</i>

Question 25 [4 marks]

4 Marks	• Students construct a fully correct user interface for use in registration, including labels for all fields
3 Marks	• Students construct a fully correct user interface for use in registration
2 Marks	• Students construct a mostly correct user interface for use in registration
1 Mark	• Students attempt to construct a user interface for use in registration

Online Registration Form

First Name

Last Name

email

Notification Method ☐ Email ☐ SMS

Create PIN

Confirm PIN

Pin must be 4 digits and not repeated

Question 26 [8 marks]

8 Marks	- Students construct a substantially correct algorithm that includes: - Correct use Python syntax - Captures all required information - Uses error checking - Uses validation - Prints all required information from the confirmation receipt - Uses sound intrinsic documentation - Interprets without error
6-7 Marks	Students construct a correct algorithm with many of the features above
4-5 Marks	Students construct a correct algorithm with some of the features above
3-2 Marks	Students construct a correct algorithm with few of the features above
1 Mark	Students construct an algorithm that shows some understanding of the problem
	<pre> 1 fname = input("Enter patient's first name: ") 2 sname = input("Enter patient's surname: ") 3 4 age = input("Enter age: ") 5 while not age.isdigit(): 6 age = input("Invalid, enter a number for age: ") 7 age = int(age) 8 9 weight = input("Enter weight (kgs): ") 10 while not weight.isdigit(): 11 weight = input("Invalid, enter a number for weight: ") 12 weight = int(weight) 13 14 surgeon = input("Enter surgeon's surname: ") 15 16 op_type = input("Type of operation (internal/external): ").lower() 17 while op_type not in ["internal", "external"]: 18 op_type = input("Invalid, enter internal or external: ") 19 20 date = input("Enter date (DD/MM/YYYY): ") 21 22 time = input("Enter time (morning/afternoon): ").lower() 23 while time not in ["morning", "afternoon"]: 24 time = input("Invalid, enter morning or afternoon: ") 25 26 booked = False 27 if booked: 28 print("Theatre not available, please reschedule.") 29 else: 30 print("\n--- Confirmation Receipt ---") 31 print("Patient:", fname, sname) 32 print("Age:", age, "Weight:", weight, "kg") 33 print("Surgeon:", surgeon) 34 print("Operation:", op_type) 35 print("Date:", date, "Time:", time) 36 print("Booking successful.") </pre>

Question 27 [3 marks]

3 Marks	• Students correctly contrast live and simulated testing
2 Marks	• Students correctly describe live and simulated testing
1 Mark	• Students identify properties of live or simulated testing
	<i>Live testing is the process of evaluating software in its actual operating environment with real users, data, and hardware. It provides accurate feedback on performance, usability, and reliability but carries risks if errors affect end-users.</i> <i>Simulated testing, by contrast, uses controlled environments or models to imitate real-world conditions without exposing the system to actual users or data. It allows developers to test safely, repeat scenarios, and detect issues early before deployment.</i>

Question 28 [3 marks]

3 Marks	• Students synthesise a fully correct SQL statement for the for the given scenario
2 Marks	• Students synthesise a somewhat correct SQL statement for the for the given scenario
1 Mark	• Students attempt to create an SQL statement
	<pre>SELECT Transaction.VehicleRegistrationNumber, Transaction.DateOfPark FROM Transaction WHERE Transaction.ParkingSpace = "1A" AND Transaction.DateOfPark <= 10/04/25 AND Transaction.CardPaymentType = "Visa" ORDERBY Transaction.DateOfPark DESCENDING</pre>

Question 29 [8 marks]

8 Marks	Students correctly assess the impact of automation on the individual, society and the environment
6-7 Marks	Students correctly assess the impact of automation on two of the three areas above
4-5 Marks	Students correctly outline the impact of automation on two of the areas above
3-2 Marks	Students correctly outline the impact of automation on one of the areas above
1 Mark	Students correctly describe automation
	<i>The impact of automation can be assessed at the level of the individual, society, and the environment, with both benefits and drawbacks evident.</i> <i>For the individual, automation often increases efficiency by removing repetitive tasks, freeing workers to focus on creative or analytical roles. For example, software engineers use automated testing to reduce manual errors and save time. However, automation can also cause job displacement, particularly in industries such as manufacturing or retail, where machines replace human labour. Individuals may require retraining or upskilling to remain employable, creating both opportunities and challenges.</i>
	<i>At the level of society, automation contributes to economic growth by enabling mass production and lowering costs. It improves access to goods and services, such as self-service checkouts and automated banking systems, which increase convenience. Yet it can also widen social inequality if certain groups lose employment while others benefit from higher productivity. The shift towards automation may also raise ethical concerns, such as the reduction of human interaction in services like healthcare or customer support.</i> <i>For the environment, automation can reduce waste and improve sustainability when systems are designed for efficiency. For instance, automated energy management in smart buildings lowers electricity consumption, while precision agriculture reduces fertiliser and water use. On the other hand, widespread automation can increase energy demand, particularly with large-scale data centres and robotics manufacturing, potentially contributing to carbon emissions if not powered by renewable sources.</i>

Question 30 [8 marks]

8 Marks	- Students construct a substantially correct algorithm that includes: - Correct use Python syntax - Captures all required information - Uses complex data structures [1D array or greater] - Uses error checking - Uses data validation or masking to capture correct values - Prints all required information regarding totals, average and highest mark - Uses sound intrinsic documentation - Interprets without error
6-7 Marks	Students construct a correct algorithm with many of the features above
4-5 Marks	Students construct a correct algorithm with some of the features above
3-2 Marks	Students construct a correct algorithm with few of the features above
1 Mark	Students construct an algorithm that shows some understanding of the problem
	<pre> 1 def capture_student_marks(student_name): 2 max_marks = [20, 20, 30, 30] 3 student_marks = [] 4 for index, max_mark in enumerate(max_marks): 5 mark = input(f"Enter mark {index+1} for {student_name} (out of {max_mark}): ") 6 while not mark.isdigit() or int(mark) < 0 or int(mark) > max_mark: 7 mark = input(f"Invalid entry, enter a number between 0 and {max_mark}: ") 8 student_marks.append(int(mark)) 9 return student_marks 10 11 student1_marks = capture_student_marks("Student 1") 12 student2_marks = capture_student_marks("Student 2") 13 14 student1_total = sum(student1_marks) 15 student2_total = sum(student2_marks) 16 17 student1_average = student1_total / len(student1_marks) 18 student2_average = student2_total / len(student2_marks) 19 20 print("\n--- Results ---") 21 print("Student 1 - Total:", student1_total, "/100 Average:", student1_average) 22 print("Student 2 - Total:", student2_total, "/100 Average:", student2_average) 23 24 if student1_total > student2_total: 25 print("Highest total achieved by: Student 1") 26 elif student2_total > student1_total: 27 print("Highest total achieved by: Student 2") 28 else: 29 print("Both students achieved the same total") </pre>

Question 31 [4 marks]

4 Marks	Students correctly recommend ways in which accessibility features can be implemented into web content to assist a wide range of users, including those with disabilities
3 Marks	Students correctly recommend ways in which accessibility features can be implemented into web content to assist users
2 Marks	Students correctly identify recommend accessibility features for users
1 Mark	Students correctly identify accessibility features
	<i>Accessibility features can be implemented into web content to support a wide range of users. Adding alt text for images allows screen readers to describe visual content to people with vision impairments. Keyboard navigation ensures that users who cannot operate a mouse can move through menus and links using only the keyboard. Providing high colour contrast and options for resizable text assists users with low vision or colour blindness. Finally, adding captions and transcripts for audio and video content supports users with hearing impairments and also benefits those in sound-sensitive environments.</i>

Question 32 [5 marks]

5 Marks	Students correctly describe CMS and justify use differences between experienced and beginner programmers based on the properties of the provided technologies
4 Marks	Students correctly describe CMS and justify use differences between experienced and beginner programmers
3 Marks	Students correctly describe CMS and identify differences between experienced and beginner programmers
2 Marks	Students correctly justify use differences between experienced and beginner programmers
1 Mark	Students describe experienced and beginner programmers
	<i>A Content Management System (CMS) is software that enables users to create, manage, and publish digital content such as websites without needing to build everything from scratch. Popular examples include WordPress, Joomla, and Drupal. CMS platforms typically provide templates, drag-and-drop interfaces, and plugin systems to extend functionality.</i> <i>For beginner programmers, CMS tools are valuable because they offer pre-designed templates, WYSIWYG (What You See Is What You Get) editors, and simple plugin installations. These features allow beginners to build functional websites quickly without requiring deep coding knowledge. The emphasis is on ease of use and reducing complexity.</i> <i>For experienced programmers, CMS tools are used differently. They can customise themes through direct editing of HTML, CSS, and JavaScript, develop custom plugins, and optimise databases for performance and scalability. Experienced developers leverage the flexibility of a CMS to integrate APIs, ensure security, and meet specific client requirements that go beyond the default tools.</i> <i>The difference arises from the properties of CMS technologies: their user-friendly interfaces support beginners in creating basic sites, while their open, extensible architecture enables experts to create highly customised, secure, and scalable solutions. This justifies why CMS platforms meet the needs of both novice and advanced users but in very different ways.</i>

